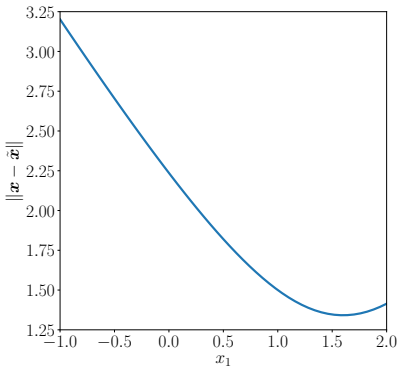
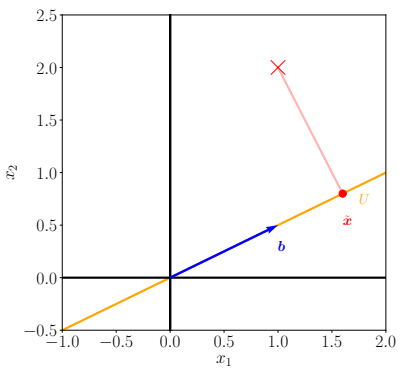


(a) Distances $\|\boldsymbol{x} - \tilde{\boldsymbol{x}}\|$ for some $\tilde{\boldsymbol{x}} = z_1\boldsymbol{b} \in U = \text{span}[\boldsymbol{b}]$; see panel (b) for the setting.



(b) The vector $\tilde{\boldsymbol{x}}$ that minimizes the distance in panel (a) is its orthogonal projection onto U. The coordinate of the projection $\tilde{\boldsymbol{x}}$ with respect to the basis vector $\boldsymbol{b}$ that spans U is the factor we need to scale $\boldsymbol{b}$ in order to "reach" $\tilde{\boldsymbol{x}}$.

Figure 10.8 Optimal projection of a vector $\boldsymbol{x} \in \mathbb{R}^2$ onto a one-dimensional subspace (continuation from Figure 10.7). (a) Distances $\|\boldsymbol{x} - \tilde{\boldsymbol{x}}\|$ for some $\tilde{\boldsymbol{x}} \in U$. (b) Orthogonal projection and optimal coordinates.

$$\frac{\partial \tilde{\boldsymbol{x}}_n}{\partial z_{in}} \overset{(10.28)}{=} \frac{\partial}{\partial z_{in}}\left(\sum_{m=1}^{M} z_{mn}\boldsymbol{b}_m\right) = \boldsymbol{b}_i \tag{10.30c}$$

for $i = 1, \ldots, M$, such that we obtain

$$\frac{\partial J_M}{\partial z_{in}} \overset{\substack{(10.30b)\\(10.30c)}}{=} -\frac{2}{N}(\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n)^\top \boldsymbol{b}_i \overset{(10.28)}{=} -\frac{2}{N}\left(\boldsymbol{x}_n - \sum_{m=1}^{M} z_{mn}\boldsymbol{b}_m\right)^\top \boldsymbol{b}_i \tag{10.31a}$$

$$\overset{\text{ONB}}{=} -\frac{2}{N}(\boldsymbol{x}_n^\top \boldsymbol{b}_i - z_{in}\boldsymbol{b}_i^\top \boldsymbol{b}_i) = -\frac{2}{N}(\boldsymbol{x}_n^\top \boldsymbol{b}_i - z_{in})\,. \tag{10.31b}$$

The coordinates of the optimal projection of $\boldsymbol{x}_n$ with respect to the basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M$ are the coordinates of the orthogonal projection of $\boldsymbol{x}_n$ onto the principal subspace.

since $\boldsymbol{b}_i^\top \boldsymbol{b}_i = 1$. Setting this partial derivative to 0 yields immediately the optimal coordinates

$$z_{in} = \boldsymbol{x}_n^\top \boldsymbol{b}_i = \boldsymbol{b}_i^\top \boldsymbol{x}_n \tag{10.32}$$

for $i = 1, \ldots, M$ and $n = 1, \ldots, N$. This means that the optimal coordinates z_{in} of the projection $\tilde{\boldsymbol{x}}_n$ are the coordinates of the orthogonal projection (see Section 3.8) of the original data point $\boldsymbol{x}_n$ onto the one-dimensional subspace that is spanned by $\boldsymbol{b}_i$. Consequently:

- The optimal linear projection $\tilde{\boldsymbol{x}}_n$ of $\boldsymbol{x}_n$ is an orthogonal projection.
- The coordinates of $\tilde{\boldsymbol{x}}_n$ with respect to the basis $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$ are the coordinates of the orthogonal projection of $\boldsymbol{x}_n$ onto the principal subspace.
- An orthogonal projection is the best linear mapping given the objective (10.29).
- The coordinates ζ_m of $\boldsymbol{x}$ in (10.26) and the coordinates z_m of $\tilde{\boldsymbol{x}}$ in (10.27)

must be identical for $m = 1, \ldots, M$ since $U^\perp = \operatorname{span}[\boldsymbol{b}_{M+1}, \ldots, \boldsymbol{b}_D]$ is the orthogonal complement (see Section 3.6) of $U = \operatorname{span}[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M]$.

Remark (Orthogonal Projections with Orthonormal Basis Vectors). Let us briefly recap orthogonal projections from Section 3.8. If $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_D)$ is an orthonormal basis of $\mathbb{R}^D$ then

$\boldsymbol{b}_j^\top \boldsymbol{x}$ is the coordinate of the orthogonal projection of $\boldsymbol{x}$ onto the subspace spanned by $\boldsymbol{b}_j$.

$$\tilde{\boldsymbol{x}} = \boldsymbol{b}_j(\boldsymbol{b}_j^\top \boldsymbol{b}_j)^{-1}\boldsymbol{b}_j^\top \boldsymbol{x} = \boldsymbol{b}_j \boldsymbol{b}_j^\top \boldsymbol{x} \in \mathbb{R}^D \tag{10.33}$$

is the orthogonal projection of $\boldsymbol{x}$ onto the subspace spanned by the jth basis vector, and $z_j = \boldsymbol{b}_j^\top \boldsymbol{x}$ is the coordinate of this projection with respect to the basis vector $\boldsymbol{b}_j$ that spans that subspace since $z_j \boldsymbol{b}_j = \tilde{\boldsymbol{x}}$. Figure 10.8(b) illustrates this setting.

More generally, if we aim to project onto an M-dimensional subspace of $\mathbb{R}^D$, we obtain the orthogonal projection of $\boldsymbol{x}$ onto the M-dimensional subspace with orthonormal basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M$ as

$$\tilde{\boldsymbol{x}} = \boldsymbol{B}(\underbrace{\boldsymbol{B}^\top \boldsymbol{B}}_{=\boldsymbol{I}})^{-1}\boldsymbol{B}^\top \boldsymbol{x} = \boldsymbol{B}\boldsymbol{B}^\top \boldsymbol{x}\,, \tag{10.34}$$

where we defined $\boldsymbol{B} := [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M] \in \mathbb{R}^{D \times M}$. The coordinates of this projection with respect to the ordered basis $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$ are $\boldsymbol{z} := \boldsymbol{B}^\top \boldsymbol{x}$ as discussed in Section 3.8.

We can think of the coordinates as a representation of the projected vector in a new coordinate system defined by $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$. Note that although $\tilde{\boldsymbol{x}} \in \mathbb{R}^D$, we only need M coordinates $z_1, \ldots, z_M$ to represent this vector; the other $D - M$ coordinates with respect to the basis vectors $(\boldsymbol{b}_{M+1}, \ldots, \boldsymbol{b}_D)$ are always 0. ♢

So far we have shown that for a given ONB we can find the optimal coordinates of $\tilde{\boldsymbol{x}}$ by an orthogonal projection onto the principal subspace. In the following, we will determine what the best basis is.

10.3.3 Finding the Basis of the Principal Subspace

To determine the basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M$ of the principal subspace, we rephrase the loss function (10.29) using the results we have so far. This will make it easier to find the basis vectors. To reformulate the loss function, we exploit our results from before and obtain

$$\tilde{\boldsymbol{x}}_n = \sum_{m=1}^{M} z_{mn}\boldsymbol{b}_m \overset{(10.32)}{=} \sum_{m=1}^{M} (\boldsymbol{x}_n^\top \boldsymbol{b}_m)\boldsymbol{b}_m\,. \tag{10.35}$$

We now exploit the symmetry of the dot product, which yields

$$\tilde{\boldsymbol{x}}_n = \left(\sum_{m=1}^{M} \boldsymbol{b}_m \boldsymbol{b}_m^\top\right)\boldsymbol{x}_n\,. \tag{10.36}$$

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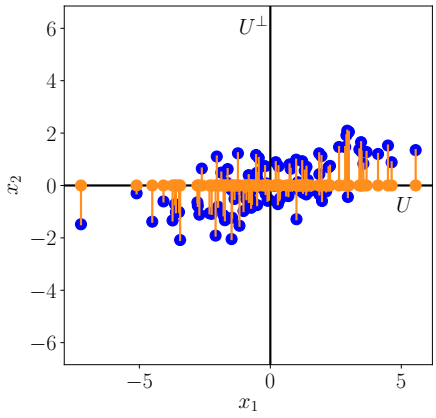


Figure 10.9 Orthogonal projection and displacement vectors. When projecting data points $\boldsymbol{x}_n$ (blue) onto subspace U_1, we obtain $\tilde{\boldsymbol{x}}_n$ (orange). The displacement vector $\tilde{\boldsymbol{x}}_n - \boldsymbol{x}_n$ lies completely in the orthogonal complement U_2 of U_1.

Since we can generally write the original data point $\boldsymbol{x}_n$ as a linear combination of all basis vectors, it holds that

$$\boldsymbol{x}_n = \sum_{d=1}^{D} z_{dn}\boldsymbol{b}_d \overset{(10.32)}{=} \sum_{d=1}^{D}(\boldsymbol{x}_n^\top \boldsymbol{b}_d)\boldsymbol{b}_d = \left(\sum_{d=1}^{D}\boldsymbol{b}_d\boldsymbol{b}_d^\top\right)\boldsymbol{x}_n \tag{10.37a}$$

$$= \left(\sum_{m=1}^{M}\boldsymbol{b}_m\boldsymbol{b}_m^\top\right)\boldsymbol{x}_n + \left(\sum_{j=M+1}^{D}\boldsymbol{b}_j\boldsymbol{b}_j^\top\right)\boldsymbol{x}_n\,, \tag{10.37b}$$

where we split the sum with D terms into a sum over M and a sum over $D-M$ terms. With this result, we find that the displacement vector $\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n$, i.e., the difference vector between the original data point and its projection, is

$$\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n = \left(\sum_{j=M+1}^{D}\boldsymbol{b}_j\boldsymbol{b}_j^\top\right)\boldsymbol{x}_n \tag{10.38a}$$

$$= \sum_{j=M+1}^{D}(\boldsymbol{x}_n^\top \boldsymbol{b}_j)\boldsymbol{b}_j\,. \tag{10.38b}$$

This means the difference is exactly the projection of the data point onto the orthogonal complement of the principal subspace: We identify the matrix $\sum_{j=M+1}^{D}\boldsymbol{b}_j\boldsymbol{b}_j^\top$ in (10.38a) as the projection matrix that performs this projection. Hence the displacement vector $\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n$ lies in the subspace that is orthogonal to the principal subspace as illustrated in Figure 10.9.

Remark (Low-Rank Approximation). In (10.38a), we saw that the projection matrix, which projects $\boldsymbol{x}$ onto $\tilde{\boldsymbol{x}}$, is given by

$$\sum_{m=1}^{M}\boldsymbol{b}_m\boldsymbol{b}_m^\top = \boldsymbol{B}\boldsymbol{B}^\top\,. \tag{10.39}$$

By construction as a sum of rank-one matrices $\boldsymbol{b}_m\boldsymbol{b}_m^\top$ we see that $\boldsymbol{B}\boldsymbol{B}^\top$ is

symmetric and has rank M. Therefore, the average squared reconstruction error can also be written as

$$\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n-\tilde{\boldsymbol{x}}_n\|^2=\frac{1}{N}\sum_{n=1}^{N}\left\|\boldsymbol{x}_n-\boldsymbol{B}\boldsymbol{B}^\top\boldsymbol{x}_n\right\|^2 \tag{10.40a}$$

$$=\frac{1}{N}\sum_{n=1}^{N}\left\|(\boldsymbol{I}-\boldsymbol{B}\boldsymbol{B}^\top)\boldsymbol{x}_n\right\|^2 . \tag{10.40b}$$

Finding orthonormal basis vectors $\boldsymbol{b}_1,\dots,\boldsymbol{b}_M$, which minimize the difference between the original data $\boldsymbol{x}_n$ and their projections $\tilde{\boldsymbol{x}}_n$, is equivalent to finding the best rank-M approximation $\boldsymbol{B}\boldsymbol{B}^\top$ of the identity matrix $\boldsymbol{I}$ (see Section 4.6). ♢

PCA finds the best rank-M approximation of the identity matrix.

Now we have all the tools to reformulate the loss function (10.29).

$$J_M=\frac{1}{N}\sum_{n=1}^{N}\|\boldsymbol{x}_n-\tilde{\boldsymbol{x}}_n\|^2\overset{(10.38b)}{=}\frac{1}{N}\sum_{n=1}^{N}\left\|\sum_{j=M+1}^{D}(\boldsymbol{b}_j^\top\boldsymbol{x}_n)\boldsymbol{b}_j\right\|^2 . \tag{10.41}$$

We now explicitly compute the squared norm and exploit the fact that the $\boldsymbol{b}_j$ form an ONB, which yields

$$J_M=\frac{1}{N}\sum_{n=1}^{N}\sum_{j=M+1}^{D}(\boldsymbol{b}_j^\top\boldsymbol{x}_n)^2=\frac{1}{N}\sum_{n=1}^{N}\sum_{j=M+1}^{D}\boldsymbol{b}_j^\top\boldsymbol{x}_n\boldsymbol{b}_j^\top\boldsymbol{x}_n \tag{10.42a}$$

$$=\frac{1}{N}\sum_{n=1}^{N}\sum_{j=M+1}^{D}\boldsymbol{b}_j^\top\boldsymbol{x}_n\boldsymbol{x}_n^\top\boldsymbol{b}_j , \tag{10.42b}$$

where we exploited the symmetry of the dot product in the last step to write $\boldsymbol{b}_j^\top\boldsymbol{x}_n=\boldsymbol{x}_n^\top\boldsymbol{b}_j$. We now swap the sums and obtain

$$J_M=\sum_{j=M+1}^{D}\boldsymbol{b}_j^\top\underbrace{\left(\frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n\boldsymbol{x}_n^\top\right)}_{=:\boldsymbol{S}}\boldsymbol{b}_j=\sum_{j=M+1}^{D}\boldsymbol{b}_j^\top\boldsymbol{S}\boldsymbol{b}_j \tag{10.43a}$$

$$=\sum_{j=M+1}^{D}\operatorname{tr}(\boldsymbol{b}_j^\top\boldsymbol{S}\boldsymbol{b}_j)=\sum_{j=M+1}^{D}\operatorname{tr}(\boldsymbol{S}\boldsymbol{b}_j\boldsymbol{b}_j^\top)=\operatorname{tr}\Big(\underbrace{\Big(\sum_{j=M+1}^{D}\boldsymbol{b}_j\boldsymbol{b}_j^\top\Big)}_{\text{projection matrix}}\boldsymbol{S}\Big), \tag{10.43b}$$

where we exploited the property that the trace operator $\operatorname{tr}(\cdot)$ (see (4.18)) is linear and invariant to cyclic permutations of its arguments. Since we assumed that our dataset is centered, i.e., $\mathbb{E}[\mathcal{X}]=\boldsymbol{0}$, we identify $\boldsymbol{S}$ as the data covariance matrix. Since the projection matrix in (10.43b) is constructed as a sum of rank-one matrices $\boldsymbol{b}_j\boldsymbol{b}_j^\top$ it itself is of rank $D-M$.

Equation (10.43a) implies that we can formulate the average squared reconstruction error equivalently as the covariance matrix of the data,

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